

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456739.77473 +/- 0.00096 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1669 +/- 0.0089
Transit depth (Rp/Rs)^2	2.78 +/- 0.3 [%]
Orbital Inclination (inc)	82.92 +/- 0.74
Airmass coefficient 1 (a1)	1.301 +/- 0.019
Airmass coefficient 2 (a2)	-0.1229 +/- 0.0034
Scatter in the residuals of the lightcurve fit is	1.47 %
Best Comparison Star	None
Optimal Aperture	3.36
Optimal Annulus	12.29
Transit Duration (day)	0.0773 +/- 0.005

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1669 $\pm$ 0.0089 vs 0.1368 (deviation 2.6 σ)
Duration fit vs published	0.0773 d vs 0.0758 d (deviation 0.23 σ)
Mid-time O-C	2.3 min
Depth SNR	14.4
Residual scatter	1.47 %

Light curve

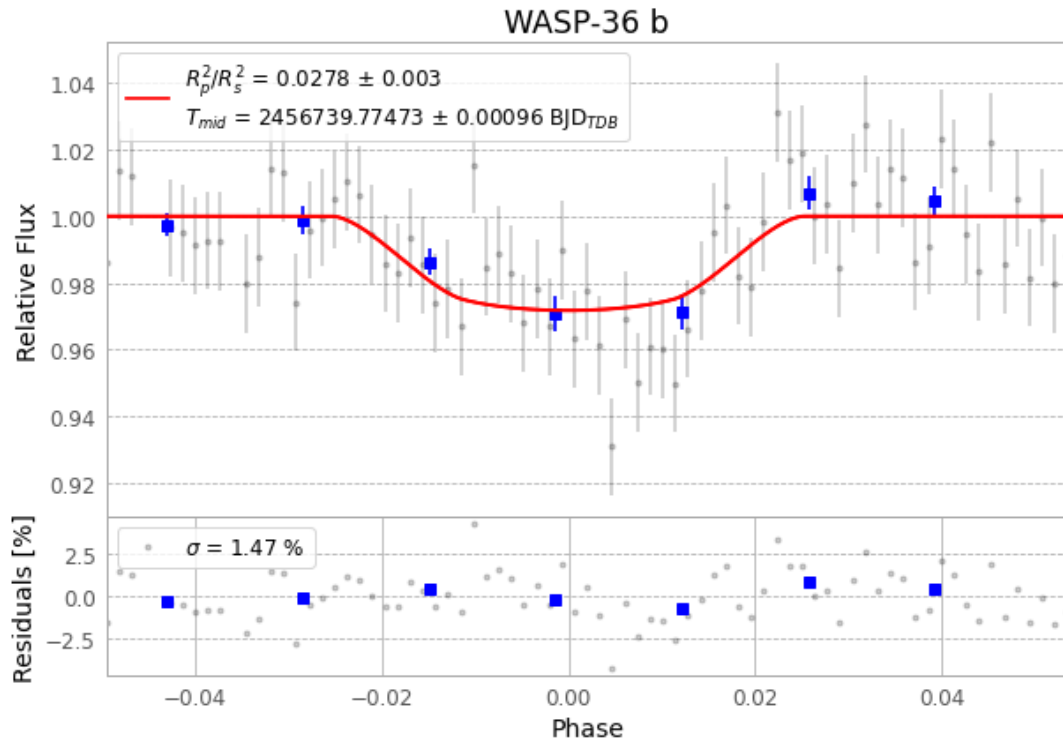


Figure 1 — FinalLightCurve_WASP-36 b_2014-03-22.png (SHA-256 93d63cf6ed948054...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [347, 248], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 edee5a5318a4952f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

77 FITS frames (51 MB), WASP-36140323043915.FITS → WASP-36140323082712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-140323.log (8e450642067b...), FinalLightCurve_WASP-36 b_2014-03-22.png (93d63cf6ed94...), FinalLightCurve_WASP-36 b_2014-03-22.csv (c2f6747d10bf...)

Compute resources

Wall time 382.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 01:25Z.